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ORIGINAL ARTICLE

# Pathological changes induced by phosphine poisoning: a study on 8 children

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## Abstract

Aluminum phosphide (ALP) has been extensively used as an economical and effective insecticide, rodenticide, and fumigant. The active ingredient of ALP is phosphine ($PH_3$), the use of which can lead to accidental inhalation and mass poisoning with high mortality. Exposure to $PH_3$ will give rise to global damage in the human body. This study reviewed 4 fatal accidents including 8 children with $PH_3$ poisoning and aimed to determine the pathological changes that resulted from exposure to $PH_3$ and, secondly, aimed to determine whether oxidative stress was involved in $PH_3$-induced neurotoxicity using histopathological and immunohistochemistry (IHC) methods. After focusing on the pathological changes on the major organs, we found severe damage induced by $PH_3$ in many systems, especially the neurological system, including neuronal, axonal, and vascular injuries as well as oxidative damage with increased expression of 4-hydroxy-2-trans-nonenal (4HNE), 8-hydroxy-2'-deoxyguanosine (8-OH-dG), and 3-nitrotyrosine (3-NT) in the brain, which indicated that oxidative stress was a crucial mechanism for neuronal death in $PH_3$ toxicity. Moreover, we observed severe myocardial and hepatocellular fatty degeneration in the tissues of the heart and liver. We considered that these characteristic changes are a suggestive sign of $PH_3$ poisoning and partly explained the toxic mechanism of $PH_3$ (inhibition of mitochondrial oxidative phosphorylation). We hope that this research could improve the understanding of the toxicity of $PH_3$ in both forensic and clinical practice.

**Keywords** Phosphine poisoning · Pathology · Oxidative stress · Neurotoxicity · Children

Yue Liang and Fang Tong contributed equally to this work.

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## Introduction

Aluminum phosphide (ALP) is a well-known insecticide, rodenticide, and fumigant with an economical price and high efficacy. It has been used to protect grain stores since the 1940s [1, 2].One problem with this grain fumigant, which has raised serious concerns, is the chemical reaction upon exposure to moisture. This process leads to the release of phosphine ($PH_3$), a lethal gas with severe toxic effects. Since an effective antidote is currently unavailable, the inhalation or ingestion of ALP or $PH_3$ contributes to an extremely high mortality rate of 70% [3, 4]. Unintentional inhalation of $PH_3$ was rarely reported before 2005, despite instances of occupational exposure having occasionally occurred [5]. Since then, due to its increasing use, there has been an alarming rise in the number of casualties associated with unintentional ALP poisoning due to its inhalation under $PH_3$ producing conditions [6–10].

$PH_3$ commonly leads to widespread organ damage and even death [3]. Some pathological changes of ALP poisoning have been observed through autopsy cases [11–13]. A study of 317 brains (21-30-year age group) associated with $PH_3$ poisoning, provided microscopic findings including neuronal degeneration, necrosis, and the disintegration of the Nissl granules, together with an eccentrically positioned, degenerated nucleus and the disappearance of the nucleolus [11]. Another study was performed on 38 fatal $PH_3$ poisonings and found the main histopathological findings were in the liver (fine cytoplasmic vacuolization of hepatocytes and sinusoidal congestion) [12]. Nonetheless, systematic studies on ALP-induced pathological changes are still limited, especially in children, who have a higher vulnerability to the poison.

Regardless, the exact toxicological mechanism of $PH_3$ is still elusive, although the suggested mechanism is considered to be the disruption of mitochondrial oxidative phosphorylation that results from an inhibition of an enzyme, cytochrome C oxidase, in mitochondria [1]. This will lead to a metabolic disorder with a reduction in ATP production. An animal study with rats has demonstrated that ALP at the $LD_{50}$ only produced an approximately 50% decrease in cytochrome C oxidase activity when compared with the control group [14]. Additionally, in studies of human tissue, the decrease in cytochrome C oxidase activity was approximately 45%, and was considered an insignificant difference between survival and fatality groups [15]. Hence, we considered that mechanisms other than the inhibition of cytochrome C oxidase should exist in explaining the toxicity of $PH_3$. The probable mechanism was hypothesized by many literature sources to include the generation of reactive oxygen species (ROS) and intracellular peroxidation, with resultant oxidative stress damage [16–18]. Nonetheless, the specific effect of oxidative stress in $PH_3$-induced neurotoxicity has not been clearly disclosed. ROS produced under oxidative stress are known to damage all cellular biomolecules (lipids, proteins, and DNA). Such biomolecules can be modified by the interaction with ROS in vivo, and some of their oxidation products are wildly used markers of oxidative stress, including the lipid peroxidation product, 4-hydroxy-2-trans-nonenal (4HNE), DNA oxidation product, 8-hydroxy-2'-deoxyguanosine (8-OH-dG), and the protein oxidation product, 3-nitrotyrosine (3-NT) [19]. This study aimed to explain the pathological changes resulting from exposure to $PH_3$ in a series of fatal accidents and, secondly, to explore whether oxidative stress was involved in the $PH_3$-induced neurotoxicity by using immunohistochemistry (IHC) of oxidative stress markers (4HNE, 8-OH-Dg, and 3-NT). The findings of this study may assist in the understanding of the general systemic damage induced by $PH_3$ as well as the role of oxidative stress in $PH_3$ toxicity.

## Materials and methods

### Information of the deceased and preparation of human samples

Samples from 4 fatal accidents of inhalational exposure to $PH_3$ used in this study were collected by Tongji Medical College, Huazhong University of Science and Technology (HUST). The information regarding the 4 accidents, involving 8 children, is listed in Table 1. The lung, liver, stomach, brain, and blood samples were obtained from the decedents, and analyzed using a headspace-gas chromatographic technique utilizing a nitrogen phosphorous detector (HS-GC/NPD) for qualitative and quantitative detection [20]. The toxicological results are listed in Table 2. The negative controls were chosen from 5 cases of children where the cause of death was due to electrocution and drowning and independent of $PH_3$ poisoning. Demographic and clinical data for all decedents are listed in Table 3. Tissue samples from the brain, heart, lung, liver, and kidneys of the 8 children were fixed in 10% formalin and paraffin blocks were made. These sections (4 μm) were used for hematoxylin and eosin (H&E) staining and IHC analysis. The tissues from the control group were sampled and tested in the same way. All experimental protocols involving human samples were approved by the victims' relatives and the Human Ethical Commission at Tongji Medical College, HUST.

### Histopathology and immunohistochemistry

Sections of the tissues from the brain, heart, lung, liver, and kidney were routinely stained with H&E. The presence of lipids in heart and liver samples was detected using Oil-red-O (ORO) staining. Nissl staining was performed on the sections prepared from brain tissue specimens. Silver staining was also performed on brain tissue to observe axonal lesions.

**Table 1** Basic information of the four accidents in the study

| Accident no. | Decedent no. | Age (years/gender) | Cause | Exposure route | Survival interval | Brief presentation | Clinical symptoms* |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 2/female | Accident | Inhalational | 26 h | A family of two slept in a bedroom where 10 bags of rice containing 15 tablets of ALP were stored. All of them presented with signs and symptoms of phosphine poisoning at 3 a.m. after 5 h sleep and were transferred to a local emergency department. Unfortunately, their disease could not be diagnosed and no special treatment was given despite frequent consultation to different physicians. They again slept in the same bedroom and were exposed to $PH_3$. The next morning, they were transported to hospital. Unfortunately, the younger child had cardiopulmonary arrest before admission and died. However, despite intensive resuscitation efforts, the older child remained hypotensive, and her clinical condition deteriorated with worsening hypoxia and metabolic acidosis. Subsequently she developed ventricular tachycardia and died 12 h after admission. | Vomiting, abdominal pain, diarrhea, convulsions, perioral cyanosis, lethargy, progressive respiratory distress and hypotension |
| | 2 | 4/female | | | 36 h | | Vomiting, abdominal pain, diarrhea, convulsions, perioral cyanosis, lethargy, progressive respiratory distress and hypotension, ECG: sinus tachycardia, complete left bundle branch block and ST-T segment changes, biological results: elevated cardiac enzymes (CPK and TnT), hypoxia and metabolic acidosis (pO2 71.1 mmHg, pCO2 22.2 mmHg, pH 7.07, bicarbonate 8.4, BE -19.9 and lactic acid 9.1 mmol/L) |
| 2 | 3<br>4<br>5 | 0.6/male<br>13/female<br>34/female | Accident | Inhalational | 10 h<br>10 h<br>12 h | A family of three (a woman with her daughter and son) slept in an upstairs bedroom. While the downstairs rooms were rented for grain storage where 300 bags of rice containing 80 tablets of ALP were stored, and on the day of accident, the ventilation system of this room was not working normally. Phosphine was released from the cracks of the ceiling into the bedroom upstairs. All of them became lethargic at 6 a.m. after 7 h sleep and were transported to a local emergency department. Children were more seriously affected, and on arrival, they experienced cardiopulmonary arrest from which they could not be resuscitated. The woman developed perioral cyanosis, respiratory distress, and went into a coma. Unfortunately, her condition continued to deteriorate and she died 3 h after admission. | Lethargy, respiratory distress, and perioral cyanosis |
| 3 | 6<br>7 | 12/female<br>13/female | Accident | Inhalational | 30 h<br>34 h | A family of two were exposed to phosphine for 5 h after ALP pellets were used to control rodents and pests in a grain storage room next to their bedroom. They began to experience abdominal pain and vomiting. They were seen by a medical physician at a local emergency department and | Vomiting, abdominal pain, perioral cyanosis, confusion and progressive decrease in blood pressure before death |

**Table 1** (continued)

| Accident no. | Decedent no. | Age (years/gender) | Cause | Exposure route | Survival interval | Brief presentation | Clinical symptoms* |
|---|---|---|---|---|---|---|---|
| | | | | | | diagnosed as experiencing food poisoning after evaluation. On the second morning after exposure, they developed a rapidly progressive decrease in blood pressure and disturbance of consciousness and were taken to hospital again, and on arrival, they experienced cardiopulmonary arrest and died. | |
| 4 | 8 | 5/Male | Accident | Inhalational | 12 h | A family of two were seen by a local medical physician with symptoms of vomiting, diarrhea, abdominal pain, and lethargy. They were dead at midnight that night, and phosphine poisoning was suspected because their bedroom next to a granary where rice was stored with ALP tablets used. Furthermore, it had been rainy for a week. | Vomiting, abdominal pain, diarrhea, and lethargy |
| | 9 | 7/Male | | | 12 h | | |

*ECG: electrocardiogram; CPK: creatine phosphokinase; TnT: troponin-T; BE: base excess.

The immunohistochemical investigation of oxidative stress damage in $PH_3$-induced neurotoxicity was performed utilizing antibodies to anti-4-HNE, anti-8-OH-dG, and anti-3-NT.

### H&E staining

In short, after de-waxing, dyeing, dehydrating, inducing transparency, and sealing, the sections were observed under a high-magnification optical microscope (× 100, × 200, × 400; Nikon, Tokyo, Japan).

### ORO staining

Frozen tissue (heart and liver) sections were rewarmed and dried, then rinsed with 60% isopropanol and stained with Oil Red O solution (Servicebio, Wuhan, China) for 8–10 min. The sections were then washed in distilled water and the nuclei were stained with Mayer's Hematoxylin. After washing with running tap water, they were mounted within a resinous medium. The lipid droplets are orange to bright red with blue nuclei.

### Nissl staining

Brain sections were stained with Nissl solution (Servicebio, Wuhan, China) for 10 min, and differentiated in Nissl differentiation solution for 4–8 s. After dehydrating in 100% alcohol, the sections were cleared using xylene and mounted within a resinous medium. Compared to the normal neurons (blue-violet in color), the injured neurons were shrunken and even disappeared, while the nuclei were darker stained.

### Silver staining

After deparaffinization and dehydration, the sections were stained with silver nitrate solution for 15 min at 37 °C in the dark and washed with tap water. The sections were then transferred into the developing liquid for at least 1 min and dried. Finally, the sections were dehydrated, hyalinized, and mounted. Nerve fibers were stained black with a golden yellow background.

### Immunohistochemical staining and semiquantitative analysis

Brain tissue sections were dewaxed and rehydrated in xylene and graded concentrations of ethanol, followed by incubation with 3% hydrogen peroxide for 10 min to eliminate endogenous peroxidase activity. Microwave-induced antigen retrieval was performed in 10 mM Tris/1 mM EDTA buffer (pH 9.0) at 95 °C for 30 min. After incubation in 5% bovine serum albumin for 20 min, to block non-specific protein binding, the sections were incubated with primary antibodies (polyclonal rabbit anti-4-HNE antibody, Abcam, UK; mouse monoclonal anti-8-OH-Dg, Abcam, UK; mouse monoclonal

**Table 2** The analytical results of total phosphine in four fatal phosphine poisoning accidents

| Decedent no. | Age (years)/gender | Phosphine results in biological samples |
|---|---|---|
| 1 | 2/female | Blood (4.65 μg/ml) |
| 2 | 4/female | Blood (6.62 μg/ml) |
| 3 | 0.6/fale | Blood (0.46 μg/mL), stomach (0.86 μg/g), liver (0.10 μg/g) |
| 4 | 13/female | Blood (0.61 μg/mL), stomach (0.86 μg/g), liver (0.17 μg/g) |
| 5 | 34/female | Blood (2.34 μg/mL), lung (0.84 μg/g), stomach (0.85 μg/g), brain (0.79 μg/g) |
| 6 | 12/female | Blood (+, without quantification), lung (+, without quantification) |
| 7 | 13/female | Blood (+, without quantification), lung (+, without quantification) |
| 8 | 5/male | Blood (3.60 μg/ml), liver (3.00 μg/g) |
| 9 | 7/male | Blood (5.28 μg/ml), lung (1.14 μg/g) |

anti-3-NT, Abcam, UK) at 4 °C overnight. Sections were then incubated for 30 min at 37 °C with HRP-conjugated secondary antibodies. Staining was performed with 3, 30-diaminobenzidine chromogen, and counterstained with hematoxylin, clarified, and mounted. Negative control was established by replacing the primary antibody with the antibody diluent. All slides were examined under a Nikon Eclipse 90i microscope (Tokyo, Japan).

To assess the degree of oxidative damage in brain tissue, a semi-quantitative score including the intensity of immunoreactivity (intensity score, IR) and the proportion of cells positively stained (PROP) was devised to measure the degree of immunoreactivity for 4-HNE, 8-OH-dG, and 3-NT. The IR ranged from 0 to 3 (0, no staining; 1, weak (+) staining; 2, moderate (++) staining; and 3, strong (+++) staining). The PROP was classified into five groups: 0, 0–4%; 1, 5–25%; 2, 26–50%; 3, 51–75%; 4, 76–100% of all cells exhibiting IR. The product of these two values was used for the calculation of the overall score (Multi-Score), as described by Harter [21]. All the measurements for every section were chosen randomly using the same image magnification (× 40, × 100, × 200, and × 400) and by the same two examiners who were blinded to the studies.

### Statistical methods

As $PH_3$ poisoning cases were rarely encountered in autopsy, the samples which we collected could not meet the standard of parameter tests. Thus, we used a non-parameter test, Wilcoxon rank sum test, for the comparisons of immunostaining profiles between the different groups. Data were analyzed using IBM SPSS Statistics 20.0 and the threshold for statistical significance was set to $p < 0.05$.

## Results

### Histological findings

**Brain** Extensive degenerated and necrotic neurons (nuclear pyknosis, karyorrhexis and karyolysis, homogenous cytoplasmic shrinkage, and red in appearance) were most prominent in the brain stem, followed by the cerebral cortex, cerebellum, thalamus, and hippocampus. In the cerebellum, Purkinje neurons showed degenerated, necrotic, even disappearing, and some of them had elongated and swollen axons. Furthermore, the parenchyma demonstrated severe edema

**Table 3** Demographic and clinical data for all post-mortem children

| | | Phosphine poisoning ($n$ = 8) | Control ($n$ = 5) |
|---|---|---|---|
| Age | Mean (range) | 7.1 years (0.6–13 years) | 7.9 years (0.4–12 years) |
| | Median | 6 years | 11 years |
| Males | | 3 (37.5%) | 2 (40%) |
| Mean survival time (range) | | 21 h (10–36 h) | / |
| Median survival time | | 19 h | / |
| Mean post-mortem delay | | 52 h | 48 h |
| Cause of death | | | |
| Phosphine poisoning | | 8 (100%) | 0 (0%) |
| Drowning | | 0 (0%) | 3 (60%) |
| Electric shock | | 0 (0%) | 2 (40%) |

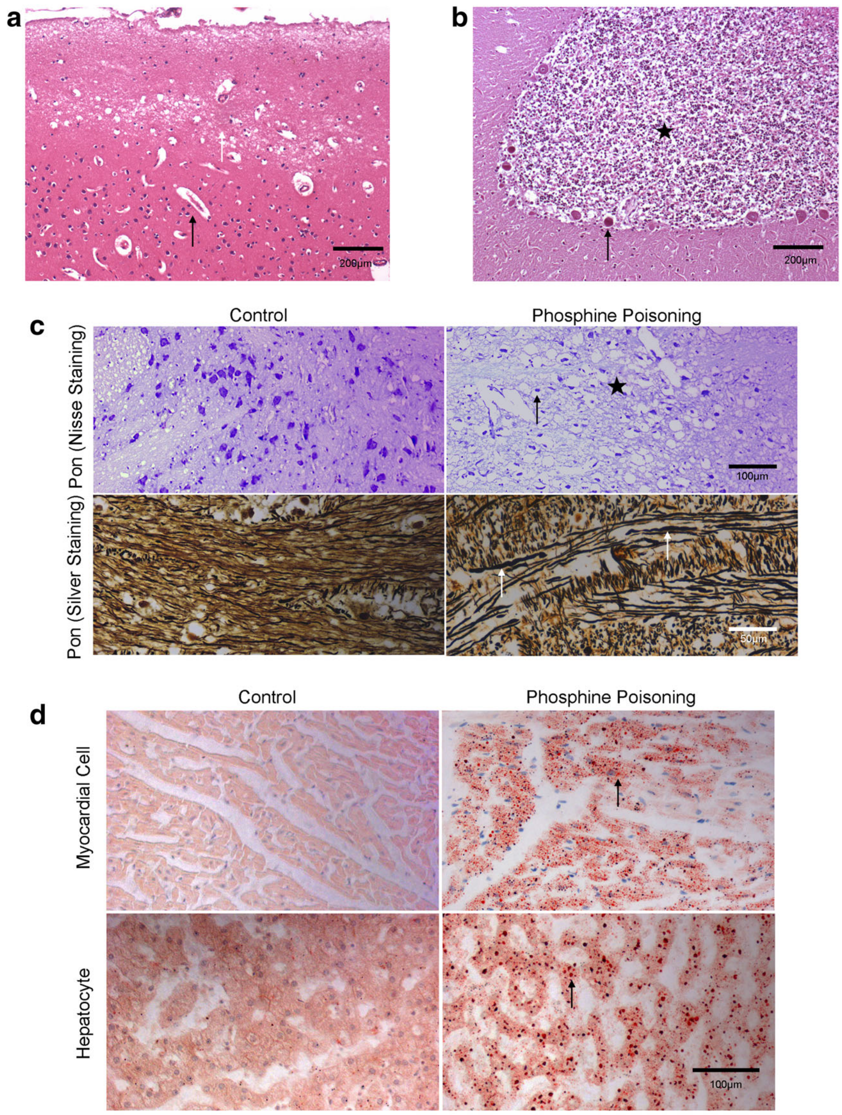


**Fig. 1** **a** Spongy regions (white arrow) and enlarged pericellular and perivascular spaces (black arrow) in superficial cerebral cortex. (H&E staining, bar = 200 μm) **b** Loose and edematous granular layer of cerebellar cortex (black asterisk), and acid stained neurons in Purkinje cell layer (black arrow) accompanied by a decrease numbers of Purkinje neurons. (H&E staining, bar = 200 μm); **c** Nissle staining and silver staining findings in the brain. Representative examples of Nissl staining in the pons: many damaged neurons (black arrow) with darker stained, shrinking, even disappeared, reduction in the number of Nissl bodies, as well as the spongy appearance in the brain tissues; relatively normal neurons with a distinct nucleus and clear cytoplasm in control. (bar = 100 μm) Representative examples of silver staining in the pons: many swollen and varicose axons in the brainstem (white arrow); smooth and closely arranged axons with uniform thickness in control. (bar = 50 μm); **d** ORO staining on liver and heart tissues. Fine cytoplasmic vacuolization of myocardial cells and hepatocytes which were identified as lipid droplets (black arrow) in the poisoned children. (bar = 100 μm)

(enlarged pericellular and perivascular spaces) and spongy appearance, predominantly in the cerebral cortex, brain stem, and granular layer of cerebellar cortex (Fig. 1a, b). Nissl staining showed that many neurons were damaged, with darker staining, shrunken, and even absent neurons being observed. A reduction in the number of Nissl bodies, as well as widening of the space between neural cells was also observed. Yet the controls exhibited relatively normal neurons with a distinct nucleus and clear cytoplasm. Representative examples of Nissl staining sections showing neural cell damage in the pons are presented in Fig. 1c. Swollen and varicose axons were observed in some regions, especially in the brainstem, and were well demonstrated by silver staining (Fig. 1c). The brain sections of the control group revealed smooth and continuous axons with a uniform thickness and close arrangement. Additionally, partial vascular walls were significantly edematous with inflammatory cell infiltration, and perivascular hemorrhage was noted in the poisoned children's sections. These changes were observed in all 8 decedents.

**Heart** We continually observed congestion, myocardial contraction band necrosis, coagulation necrosis, vacuolar degeneration, edema, and scattered inflammatory cells in the heart tissues. Fine cytoplasmic vacuolization of myocardial cells in some regions, which was best presented with ORO staining, was observed in all the poisoned children's sections but not in the control group (Fig. 1d). Occasional sub-epicardial hemorrhage with inflammatory cells infiltration was found in accident 1 and accident 2.

**Lung** Severe edema and moderate hemorrhaging was present in many alveoli. Collapsed alveoli and bronchioles and alveolar dilatation were observed in many regions. Necrosis and exfoliation of the bronchial mucosa epithelium, with peribronchiolar inflammatory infiltration, was occasionally observed. Prominent vascular congestion and scattered hemorrhaging in the interstitium was also observed.

**Liver** Focal hepatocyte necrosis with inflammatory cell infiltration was observed. Some hepatocytes showed mild edema. Fine cytoplasmic vacuolization of the hepatocytes in many regions which were identified as being lipid droplets by using ORO staining in most areas (Fig. 1d). Portal inflammation, central vein, and sinusoidal congestion were observed in most accidents. Occasional bile stasis was found in decedent 1.

**Kidney** Epithelial necrosis was observed in both the proximal and distal renal tubules in all eight decedents. The cytoplasm of these tubular epithelial cells was clearly stained red. Some cellular or granular casts formed by necrotic and exfoliated tubular epithelial cells were occasionally found. Tubular dilation, edema, and even balloon degeneration were occasionally observed. Interstitial hemorrhage with inflammatory cell infiltration was found in decedent 1.

Numerous internal organs, including the laryngeal and tracheal mucosa, were congested with petechial hemorrhages. Additionally, in decedents 3 and 4, small bubbles were observed in some blood vessels of several organs, mainly including the lung, followed by the heart and brain. The primary histologic findings of these accidents are listed in Table 5.

## Oxidative damage in the brain

The lipid peroxidation level was determined by evaluating 4-HNE immunoreactivity; the 8-OH-dG level indicated the degree of oxidative damage to DNA, and protein oxidative damage was detected by assessing the 3-NT level. As shown in Fig. 2, cerebral 4-HNE, 8-OH-dG, and 3-NT immunoreactivity was significantly higher in the poisoning poisoned children when compared to the children of the control children group. (4-HNE: $PH_3$ poisoning group 5.57 ± 0.72 vs. control group 1.87 ± 0.13, $p$ < 0.05; 8-OH-dG: $PH_3$ poisoning group 11.42 ± 0.49 vs. control group 4.55 ± 1.25, $p < 0.05$; 3-NT: $PH_3$ poisoning group 7.25 ± 0.68 vs. control group 1.85 ± 0.45, $p < 0.05$, Fig. 2, Table 4). 4-HNE and 3-NT expression was mainly distributed in the cytoplasm of neurocytes in all brain regions, particularly in the cerebral cortex and the brainstem, while 8-OH-dG was mainly expressed in the nucleus of neurocytes. It indicated significant lipid peroxidation, protein, and DNA oxidative damage in the brain during the poisoning process.

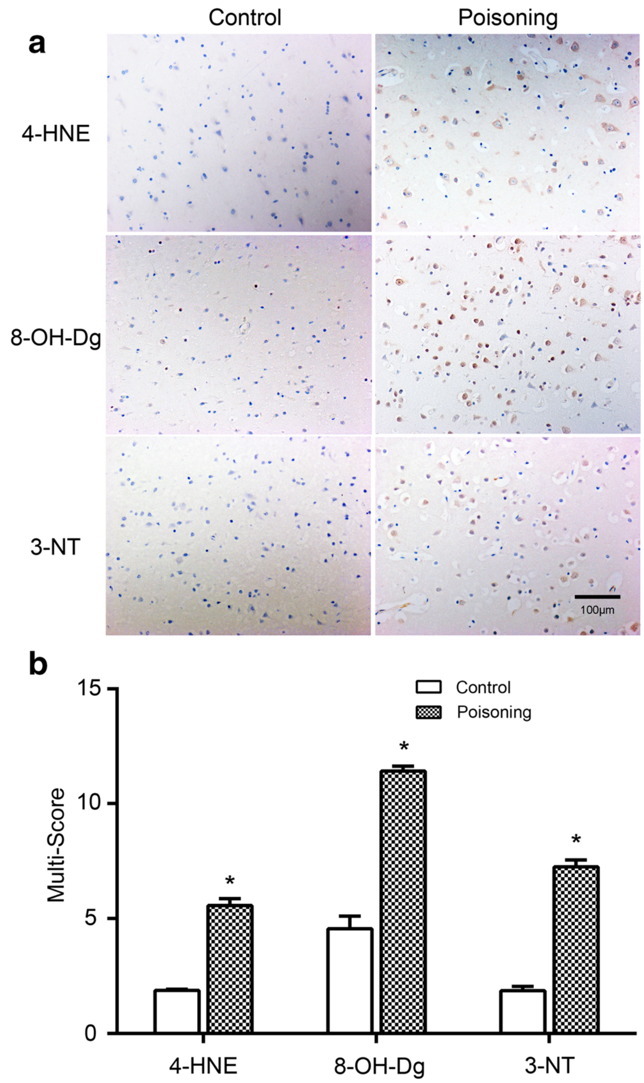


**Fig. 2** 4-HNE, 8-OH-Dg, and 3-NT immunoreactivity in the brain: **a** 4-HNE, 8-OH-dG, and 3-NT expression in neurons in the cerebral cortex (bar = 100 μm) **b** Semiquantitative analysis result of 4-HNE, 8-OH-dG, and 3-NT (*$p$ < 0.05).

## Discussion

To our knowledge, reports of $PH_3$ or ALP poisoning are well documented in the literature. We searched PubMed and Google Scholar for all years and all languages using the key words “($PH_3$ OR ALP) poisoning AND (death OR autopsy).” As this research was focused on the pathological changes on the major organs, only fatal cases caused by $PH_3$ or ALP poisoning with autopsy and pathology findings were included (Supplementary table). Then we compared our study with these 20 reports on pathological findings in the main organs (Table 5) [8, 9, 11, 12, 22–37]. In our review of 20 reports, the age reported ranged from 0.4 to 66 years old and the mean age was approximately 31.1 years old whereas all the decedents in the present study were children of 0.6~13 years old. To the best of our knowledge, children generally do not have age-related diseases (e.g., CHD, cerebral infarction) and lifestyle dependent damage (e.g., smoking, alcohol use) which tend to affect the examination results. In addition, according to available literatures [10, 24, 38] and accident 2 here (two children

**Table 4** Semiquantitative analysis results of oxidative stress markers immunoreactivity in the brain

| Oxidative stress markers | Phosphine poisoning | | Control | |
|---|---|---|---|---|
| | Intensity score, IR | Multi-score | Intensity score, IR | Multi-score |
| 4-HNE | (++)-(+++) | 5.57 ± 0.72 | (+) | 1.87 ± 0.13 |
| 8-OH-dG | (+++) | 11.42 ± 0.49 | (+)-(+++) | 4.55 ± 1.25 |
| 3-NT | (++)-(+++) | 7.25 ± 0.68 | (+)-(++) | 1.85 ± 0.45 |

died earlier and the blood PH3 concentration was much lower than their mother's), it is suggested that $PH_3$ is more harmful to children than adults. Therefore, we consider that by obtaining samples from child decedents of $PH_3$ poisoning, we could eliminate the unrelated pathological changes seen in adult samples described above and deliver a more accurate pathological analysis of $PH_3$ poisoning.

As a highly toxic respiratory poison, $PH_3$ affects all major organ systems [3]. Cardiovascular toxicity is highly varied and extensive, manifesting with ventricular arrhythmias, circulatory failure, and shock, and it is considered to be the most common and most important cause of mortality [7, 10, 30]. Our research included these manifestations and was consistent with previous literatures. In accident 1, the ECG of decedent 2 demonstrated abnormal cardiovascular manifestations including tachycardia, complete left bundle branch block, and ST-T segment change. According to previous studies, these findings were considered to be suggestive signs contributing to fatality in $PH_3$ poisoning [39]. Cardiovascular pathological findings of these reports also revealed severe myocardial damages as was demonstrated in Table 5. In addition to the lesions reported in the literature, we also found mild steatosis of myocardial cells which suggested a process of fatty acid metabolic disorder induced by $PH_3$ toxicity.

In our study, in addition to cardiovascular toxicity, the exposed children also presented with clinical manifestations of neurological toxicity including convulsions, confusion and lethargy, as well as significant neuropathological changes which were obvious and should be considered crucial to the toxic process of $PH_3$. Though several studies occasionally mentioned brain damage [11, 22–25], only one research article using human brains (21-30 year age group) has systematically studied pathological changes in the brain associated with $PH_3$ poisoning, including neuronal degeneration, necrosis together with axonal degeneration in cerebral cortex, cerebellar cortex, and subcortical zone [11]. Similar changes were observed in our study and more extensively than in the above research, including the cerebral cortex, cerebellum, thalamus, hippocampus, and white matter, which suggested diffuse and severe neurotoxicity induced by $PH_3$ in children. And beyond that, we found pathological features of axonal injury (swollen and varicose) in some brain regions using silver staining. Moreover, severe brain edema manifested as enlarged pericellular and perivascular spaces predominantly in the white matter, and edematous vascular walls and perivascular hemorrhage were also observed, indicating the occurrence of vasogenic brain edema and the breakdown of blood–brain barrier (BBB) [40, 41]. We considered that the previous research did not intentionally ignore the CNS damage and that the absence of reported brain damage may be ascribed to the original samples taken. Most researchers collected tissue samples from the decedents in a wide age range whereas all the decedents in the present study were children of 0.6~13 years old. Research has revealed that the developing brain consumes more energy than that of adults, and oxidative phosphorylation is the primary source of ATP, providing most of the energy used to support life [42, 43]. Under the premise that $PH_3$ poisoning leads to oxidative phosphorylation inhibition, this may explain the significant brain damage in our study, where the decedents were all children.

Since inhalational exposure is thought to be the primary route of intoxication in $PH_3$ poisoning, the respiratory system is a specific location that is vulnerable to $PH_3$. The decedents in our study exhibited clinical features of respiratory toxicity as well as severe pathological changes, such as extensive edema and hemorrhage, as found in other literature [8, 22, 23, 25–29, 31–34] (Table 5). Though pneumorrhagia is an unspecific and commonly observed finding during autopsy, cases of acute $PH_3$ poisoning due to inhalational exposure are usually accompanied by severe pneumorrhagia. Furthermore, small bubbles were found in some pulmonary interstitial vessels of two children in our study, indicating the destruction of the gas and blood barrier induced by the vessels and alveolar wall injury. It is known that the renal, hepatobiliary, and hematological systems are also affected in $PH_3$ poisoning [44]. It is worth noting that vacuolar degeneration of the hepatocytes is one of the most common types of hepatic damage in $PH_3$ poisoning noted in other literature [12, 31]. In our study, some fine cytoplasmic vacuolization of the hepatocytes were identified as being lipid droplets by using ORO staining, which was more suggestive of hepatocyte fatty acid metabolic disorder than other literature due to the fact that children generally do not have the basic lesions such as fatty liver that often occur in adults.

Previous studies considered that the predominant mechanism of $PH_3$ toxicity was ATP depletion resulting from the inhibition of mitochondrial oxidative phosphorylation [1, 13]. This may lead to widespread organ damage, as observed in our study. We also observed myocardial and hepatocellular

**Table 5** Comparison of pathological findings in main organs in the present study with the previous literature on fatal $PH_3$ poisoning

| $PH_3$ toxicity | Present study mean age (range) = 7.1 (0.6–13 years) | Previous literature mean age (range) ≈ 31.1 0.4–66 years)* |
|---|---|---|
| Neurological | | |
| 1) Edema | 100% (8/8) | Fang et al. [22], Tripathi et al. [11], Musshoff et al. [23], Yan et al.[24] |
| 2) Extensive neuronal degeneration, necrosis, even disappearance | 100% (8/8) | Fang et al. [22], Sinha et al. [25], Tripathi et al. [11] |
| 3) Axonal degeneration (swollen and varicose) | 100% (8/8) | Tripathi et al. [11] |
| 4) Edematous vascular walls | 50% (4/8) | — |
| 5) Perivascular hemorrhage | 100% (8/8) | — |
| 6) Oxidative stress injury | 100% (8/8) | — |
| Cardiovascular | | |
| 1) Contraction band necrosis/coagulation necrosis | 100% (8/8) | Jadhav et al. [26], Abder-Rahman et al. [27], Jain et al. [28], Memiş et al. [29], Musshoff et al. [23] |
| 2) Vacuolar degeneration | 100% (8/8) | Shah et al. [30] |
| 3) Mild steatosis | 100% (8/8) | — |
| 4) Hemorrhage | 25% (2/8) | Abder-Rahman et al. [27], Abder-Rahman [8], Jadhav et al. [26], |
| 5) Inflammation infiltration | 100% (8/8) | Abder-Rahman [8], Jain et al. [28], Yan et al.[24] |
| Respiratory | | |
| 1) Edema | 100% (8/8) | Misra et al [31], Abder-Rahman et al. [27], Fang et al. [22], Sinha et al. [25], Jain et al. [28], Memiş et al. [29], Yadav et al. [32], Musshoff et al. [23], Abder-Rahman [8], Hugar et al. [33], Meena et al. [34] |
| 1) Collapse of alveoli | 100% (8/8) | Sinha et al. [25] |
| 2) Necrosis and exfoliation of the bronchial mucosa epithelium | 62.5% (5/8) | Misra et al [31] |
| 3) Peribronchiolar inflammatory infiltration | 62.5% (5/8) | Jain et al. [28] |
| 4) Hemorrhage | 100% (8/8) | Abder-Rahman et al. [27], Fang et al. [22], Sinha et al. [25], Musshoff et al. [23], Abder-Rahman [8], Jadhav et al. [26], Hugar et al. [33] |
| Hepatic: | | |
| 1) Vacuolar degeneration | 100% (8/8) | Misra et al [31], Saleki et al. [12] |
| 2) Steatosis | 100% (8/8) | Fang et al. [22], Saleki et al. [12], Sinha et al. [25] |
| 3) Portal inflammation | 100% (8/8) | Fang et al. [22], Saleki et al. [12], Shadnia et al. [35] |
| 4) Hepatocyte necrosis | 100% (8/8) | Singh et al. [36], Saleki et al. [12], Musshoff et al. [23], Jadhav et al. [26], Sinha et al. [25], Hugar et al. [33] |
| 5) Bile stasis | 12.5% (1/8) | Sinha et al. [25] |
| Renal | | |
| 1) Degeneration and necrosis of tubular epithelium | 100% (8/8) | Sinha et al. [25], Jain et al. [28] |
| 2) Cellular or granular casts | 100% (8/8) | — |
| 3) Interstitial infiltration of inflammatory cells | 12.5% (1/8) | — |
| Systemic | | |
| 1) Congested | 100% (8/8) | Sinha et al. [25], Misra et al [31], Abder-Rahmanet al. [27], Anger et al. [37], Fang et al. [22], Jain et al. [28], Jadhav et al. [26], Behera at al. [9], Meena et al. [34], Yan et al.[24] |
| 2) Small bubbles in the blood vessels (lung, heart, brain) | 25% (2/8) | — |

*"—" means it has not been reported in literature

**Table 6** Review and comparison of the oxidative stress damage of $PH_3$ poisoning in the present study with the previous literature

| Authors and publication year | Species/age | Exposure route | Samples | Oxidative stress markers * |
|---|---|---|---|---|
| Chugh et al. (1996) [45] | Patients/ - ($n$ = 45) | ALP poisoning | Serum | SOD↑, CAT↓, MDA↑ |
| Hsu et al. (1998) [46] | Hepa 1c1c7 | Saturated aqueous solution of PH3 | Liver cells | ROS↑, MDA↑, 4HNE↑, 8-OH-dG↑ |
| Hsu et al. (2000) [47] | Wistar rats/250 ± 30 g | ALP solution, i.p. | Brain, liver, lung | GSH↓, GSSG↑, MDA↑, 4HNE↑, 8-OH-dG↑ |
| Hsu et al.(2002) [48] | Wistar rats/250 ± 30 g | ALP solution, i.p. | Kidney, heart | GSH↓, MDA↑, 4HNE↑, 8-OH-dG↑, GSH-Px↓, SOD↑, CAT↓ |
| Anand et al. (2012) [17] | Wistar rats/140–200 g | ALP, orally | Liver, brain | CAT↓, MDA↑ |
| Anand et al. (2013) [16] | Patients/ mean age = 28.8 ($n$ = 21) | ALP ingestion | Platelet, plasma | CAT↓, TTM↓, MDA↑, protein carbonyl content↑ |
| Agarwal et al. (2014) [49] | Patients/mean age = 27.74 ± 8.86 ($n$ = 24) | ALP poisoning | Serum | SOD↓, CAT↓ |
| Liu et al. (2015) [50] | Mutant drosophila strainW$^{1118}$ | $PH_3$ fumigation treatment | Insects | $H2O_2$↑, LPO↑, SOD1↑, $SOD_2$↑, CAT↓ |
| Abdolghaffari et al. (2015) [51] | Wistar rats/200–250 g | ALP, gavage | Herat | FRAP↓, MDA↑, SOD↑, ROS↑, TTM↓ |
| Solgi et al. (2015) [52] | Wistar rats | ALP, gavage | Plasma | FRAP↓, MDA↑ |
| Jafari et al. (2015) [53] | Wistar rats/ 200–250 g | ALP, gavage | Heart | CAT↓, MDA↑, TTM↓, ROS↑ |
| Baghaei et al. (2016) [18] | Wistar rats/ 200–220 g | ALP, gavage | Heart, plasma | ROSrt CATrt free plasma irone |
| Maleki et al. (2019) [54] | Wistar rats/8 weeks old | ALP, gavage | Heart | ROS↑, MDA↑, GSH↓ |
| Aminjan et al. (2019) [55] | Wistar rats/220 ± 20 g | ALP, gavage | Liver | SOD↓, ROS↑, CAT↓, GSH↓, G6PD↓, MPO↑ |
| The present study | Children/0.6~13 ($n$ = 8) | $PH_3$, inhalation | Brain | 4HNE↑, 8-OH-dG↑, 3-NT↑ |

**SOD*, superoxide dismutase; *CAT*, catalase; *MDA*, malondialdehyde; *GSH*, glutathione; *GSSG*, GSH disulfide; *4HNE*, 4-hydroxy-2-trans-nonenal; *8-OH-dG*, 8-hydroxydeoxyguanosine; *GSH-Px*, GSH peroxidase; *ROS*, reactive oxygen species; *FRAP*, ferric reducing/antioxidant power; *TTM*, total thiol molecules; *G6PD*, glucose 6 phosphate dehydrogenase; *MPO*, myeloperoxidas

steatosis. This phenomenon can be a specific feature of fatty acid metabolic disorder induced by the absence of ATP. These findings supported the above toxic mechanism of PH3. Additionally, the role of oxidative stress has been emphasized in numerous studies when evaluating the toxic mechanism of $PH_3$ (Table 6) [16–18, 45–55]. We found that most studies in our review (Table 6) have demonstrated oxidative stress damage in rats (64.3%) and the samples mainly involved the heart and plasma [17, 18, 47, 48, 51–55]. These studies mainly found the significant inhibition of the defense system, including nonenzymatic molecules (GSH and TTM) as well as enzymatic scavengers of ROS (SOD, CAT, GSH, and GSH-Px). They also found elevated lipid peroxidation products (MDA and 4HNE) and oxidative damage to DNA in response to elevated 8-OH-dG. A few studies of human $PH_3$ poisoning in our review revealed that antioxidant enzyme (CAT and SOD) activity was decreased in serum ALP poisoning patients, resulting in lipid peroxidation [16, 49]. Nevertheless, it was previously unclear whether oxidative stress was initiated in the neurotoxic mechanisms of $PH_3$, especially in children who have a higher vulnerability to oxidative stress [56]. Thus, in our study, we aimed to determine whether oxidative stress was involved in $PH_3$-induced neurotoxicity.

As with the other studies in Table 6 [47, 48], we also applied 4-HNE to investigate oxidative damage of lipid and 8-OH-dG to investigate oxidative stress damage in DNA. As one of the major products of lipid peroxidation, 4HNE is considered a biomarker of oxidative stress and is identified as one of the most abundant and active lipid peroxides [57, 58]. Furthermore, 4-HNE has been shown to induce cell death in specific contexts [59]. Semiquantitative analysis results demonstrated that the immunoreactivity of 4-HNE was significantly higher in the poisoned children than in the control groups, and 8-OH-dG was widely and strongly expressed in the nucleus of neurocytes ($p <$ 0.05). This result indicated that lipid peroxidation was enhanced, and severe oxidative damage to DNA in the brain occurred after $PH_3$ poisoning. On the other hand, evidence suggests that proteins are major targets for ROS in oxidative stress situations in vivo and may often be more important than damage to lipids [60, 61]. 3-NT, as a protein oxidation product, is thought to be a stable marker of protein oxidation [62]. In this regard, we added 3-NT in IHC staining to detect oxidative damage of proteins in our study, and the results showed 3-NT level was significantly elevated in the brain after $PH_3$ poisoning. Therefore, it is suggested that $PH_3$ could lead to the oxidative stress, oxidative damage of lipids, proteins, and DNA, and ultimately extensive neuronal death.

## Conclusion

In summary, our study revealed that acute $PH_3$ poisoning results in widespread organ damage with significant pathological changes in the major organs. As for neurotoxicity induced by $PH_3$, we found severe brain damage including neuronal, axonal, and vascular injury as well as significantly increased expressions of 4HNE, 8-OH-dG, and 3-NT in the brain, indicating that oxidative stress plays a major role in PH3 poisoning pathogenesis. Moreover, the myocardial and hepatocellular fatty degeneration we observed can be considered characteristic pathological changes of $PH_3$ poisoning and partly explained the toxic mechanism of PH3 (inhibition of mitochondrial oxidative phosphorylation). Considered together, this data may improve the understanding of the toxicity of $PH_3$ and provide clues for possible interventions in acute $PH_3$ poisoning.

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## Compliance with ethical standards

**Conflict of interest** The authors declare that they have no conflicts of interest.

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